Primal, 1st-order, Single Variable

$F(x) = f(x) + g(x)$

$$
\begin{tikzcd}
x = \operatorname{argmin} F(x) \arrow[r, "\text{Linear}"] \arrow[d, "+\frac{L}{2}\|\cdot\|^2"] & \text{Grad Descend} \arrow[d, "+\frac{L}{2}\|\cdot\|^2"] \arrow[dr] & \\
\text{Proximal Method} \arrow[r, "\text{Linear}"] \arrow[dr, "g = I_C(x)"] & \text{Prox Grad} \arrow[dr] & \\
 & \text{Proj}_C(x^k) & \text{Proj Subgrad}
\end{tikzcd}
$$

Nonsmooth f

Subgrad

$\min\ f(x) + g(x)$ f 表示 cost, g 为 penalty e.g. LASSO

	$f = 0$	f 光滑	f 非光滑
无 g	$\times$	Grad	Sub Grad Ch3, Ch8
$g = I_C(x)$	Proj Ch6	Proj Grad	Proj SubGrad Ch8
g 一般	Prox Ch6	Prox Grad Ch10	...

Multi-block

$x, y = \operatorname{argmin} F(x, y)$

$$
\begin{tikzcd}
\text{Jaccobi / Gauss-Seidel} \arrow[r, "\text{Linear}"] \arrow[d, "+\frac{L}{2}\|\cdot\|^2"] & \text{BCD} \arrow[d, "+\frac{L}{2}\|\cdot\|^2"] \\
\text{Proximal GS} \arrow[r, "\text{Linear}"] & \text{PALM}
\end{tikzcd}
$$

Dual

Grad Descend $\rightarrow$ Dual Ascent $\xrightarrow{+\frac{\rho}{2}\|\cdot\|^2}$ ALM

Proj Subgrad $\rightarrow$ Frank-Wolfe

Multi-block ADMM $\xrightarrow{+\frac{L}{2}\|\cdot\|^2}$

$$
\begin{cases}
\text{Jaccobi}: \text{PMM (Proximal Method of Multiplier)} \\
\text{GS}: \text{Proximal ADMM}
\end{cases}
$$